

Business Process Management System Implementation Model for Improving Employee Performance

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Abstract— Business Process Management Systems focuses on the development of the company's business processes in improving the relationship between the customer and the supplier (supply chain management) to fulfill customer demand. This process focuses on understanding the needs of the company's business processes to improve employee performance. The purpose of the study was to define business process management systems implementation model that can be used to measure employee performance improvement. The research methodology used the literature study, observations, questionnaires, and analysis of data using factor analysis. Results obtained from the research is a new factor analysis and indicators that can be used as a basis business process management systems implementation model for employee performance measurement model. Thus it can be concluded that in this study, the example of the company used in the study has an inadequate business process management system so that the results of this study produce a business process management implementation model that can be used as a benchmark in improving employee performance in a better direction.

Keywords—business process management system, key performance indikator, factor analysis

I. INTRODUCTION

Each company has a goal to be able to distribute or provide products and services that can meet the needs of the environment (customers). Therefore, the company will not be able to function without the resources provided by the environment (suppliers). [17] The second element above has their respective roles in which suppliers or vendors act usually referring to the supply of raw materials, machines, services, people, and information used by businesses to produce products and services; whereas customers are parties who use marketed products and services. The flow of physical resources from suppliers to companies then

to customers is called a supply chain that needs to be managed properly so that the flow occurs in a timely and efficient manner. CV. Macro Komputrindo is one company that is engaged in the sale and distributor of equipment or computer hardware. Because of the high demand from customers, and the amount of inventory owned by the company can sometimes not meet customer demand, giving rise to various problems such as information and data that are not integrated and there are many business processes carried out manually such as recording and reporting goods in the warehouse is still done manually and still has differences with employee-managed reporting and records and inventory that cannot meet customer demands. This causes business processes in the company to be ineffective and inefficient.

In the current business development, organizations become more aware of the interests of business processes in terms of effectiveness, efficiency and compliance. And as a result more and more organizations are paying attention and focusing on operational aspects of BPM and functional [19]. For companies, information related to SCM (Supply Chain Management) is an important asset because the Supply Chain consists of all parties involved directly or indirectly in meeting customer demand [5]. In order for business processes at CV. Macro Komputrindo can run well, CV. Macro Komputrindo created a Business Process Management System (BPMS) to manage business information on CV. Komputrindo Makro is known as MKIIS (Macro Komputrindo Internal Information System), which serves as a means of sharing information and accessing information quickly to facilitate tasks and improve employee performance. Through the use of MKIIS, CV. Komputrindo Makro hopes that all employees can use and implement BPMS effectively and can contribute to the company by improving employee performance and customer satisfaction. Therefore, it is necessary to evaluate

the implementation of BPMS at this time to find out what factors and indicators can be used as a business management process implementation model that can affect employee performance improvement.

II. METHODOLOGY

The methodology that will be used in the writing of the thesis is as follows:

1. Methods of data collection :

In carrying out this study, to obtain the data and information needed is obtained by:

a. Library research methods such as literature study by studying and reading literature, scientific journals and other media to obtain the necessary information related to the research.

b. Observation, in which the authors make direct observation of the business processes that are running on the company. From the observations were obtained, the author will make development for business processes.

c. The questionnaire, containing a number of questions to be asked either to the management and employees to obtain the information needed in the analysis process.

d. Interview, in which the author conducted a question and answer session directly or through an e-mail to interested parties in this study.

2. The research method

The research method used is by using descriptive analysis and factor analysis to test the validity and reliability of data. The several stages of research conducted:

a. The First Stage

Determining factors and indicators of what will be used to evaluate BPMS. In this study use traditional factors consist of: BPM [1], SCM Flow [12], Characteristics of SCM [10], and BSC (Balanced Scorecard) [9]. Indicator is obtained which will be used to get the value in the search for information.

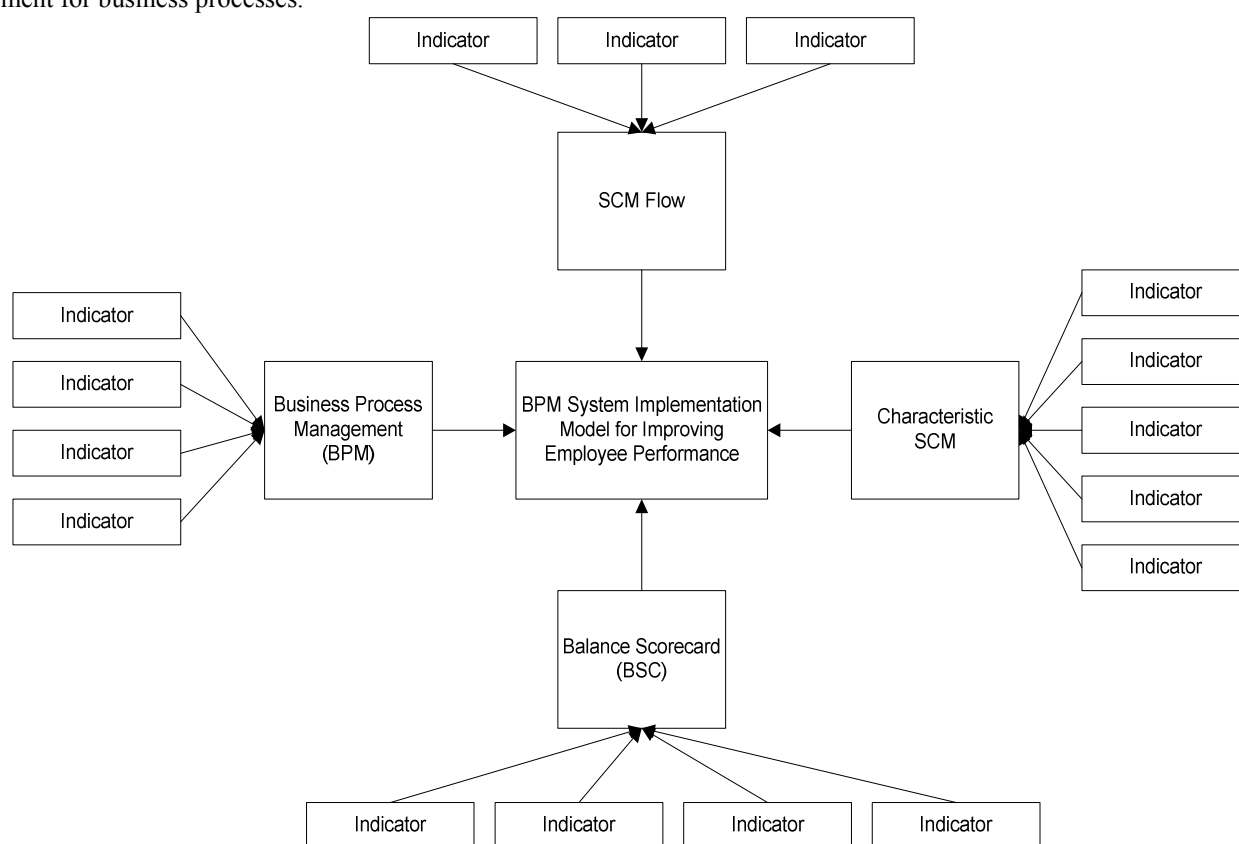


Fig. 1. Research Model Concept

TABLE I. MATRIX RELATIONSHIP BETWEEN THEORY / CONCEPT INDICATOR FACTOR

Factor	Indicator	Source
Business Process Management	Target Management	Balaban [1]
	Performance Management	Balaban [1]
	Resources Management	Balaban [1]
	Process Interface Management	Balaban [1]

Supply Chain Management Flows	Information Flow	Vanpoucke [24]
	Goods Flow	James [12]
	Financial Flow	Whitten [27]
Characteristic SCM	Production	Feng [10]
	Inventory	Kifor [13]
	Location	Datta [7]
	Transportation	Storey [22]
	Information	James [12]
Balance Scorecard	Financial	Ervil [9]
	Customer	Ervil [9]
	Internal Business	Wasposito [25]
	Learning and Growth	Wasposito [25]

b. The Second Stage

Distributing questionnaires and data collection has been given by the respondent, after which the data were analyzed with the aim of getting new indicators and factors that may affect employee performance improvement CV. Macro Komputrindo.

c. The Third Stage

After getting the new factors that can affect the performance of the employees, from the results of the analysis of made a conclusion that will be used as a suggestion to the CV. Macro Komputrindo to improve the quality of BPMS in order to improve employee performance CV. Macro Komputrindo.

III. RESULT AND DISCUSSION

From the results of factor analysis indicators can know anything that can affect employee performance improvement CV. Macro Komputrindo.

The first new factor, consisting of indicator:

1. IB2= CV. Macro Komputrindo always strives to update products to meet demand and create customer satisfaction.
2. LG2= A company always strives to provide a means of supporting and extremely adequate equipment.
3. LG4= Employees always want to know how to make progress when doing a job.
4. LG3= I always try to obey all rules of the company.
5. LG1= I ever received a reward or award from the company for his accomplishments I have achieved.
6. IB1= company is always seeking employees to always respond to all customer complaints effectively and efficiently.
7. Cust1= I can help the company to improve the customer satisfaction with the good service and accurate information.
8. Cust2= I always try to provide good service to customers.
9. Fin1= The application or technology used in the enterprise can increase profit / profit for the company.

10. LG5= A company has always tried to accept input, criticism and suggestions from employees.

11. PIM1= The existence of the joint meeting plan targets what process needs to be achieved in order to be appropriately structured, rational, logical, and relevant to setting goals - goals.

The second new factor, consisting of the following indicators:

1. Cust2= I always try to provide good service to customers.
2. PIM1= The existence of the joint meeting plan targets what process needs to be achieved in order to be appropriately structured, rational, logical, and relevant to setting goals - goals.
3. Loc1= I always try to check the goods delivered in accordance with demand, good condition, and delivered on time.
4. Trans1= The company always strives to ensure the delivery of goods to customers is done on time, and up to good condition.
5. Inf1= I have always tried to provide information related to the items required by the customer.
6. Prod1= All of the staff have easy to access the information required in relation to the needs of our work.
7. Inv1= I often have difficulty in searching the data / documents in previous years.
8. GF1= I can provide information related to the availability of products are up to date with the adequate facilities.
9. IF1= Company provide adequate facilities so as to provide effective and efficient services.

The third new factor, consisting of the following indicators:

1. PM1= Employees are given the opportunity to submit suggestions / opinions on the company's decision.
2. RM1= The company needs the support of human resources for achieving the goals.
3. Tar1= All employees are aware of the objectives / targets to be achieved by the company.

4.LG6= Good communication between superiors and subordinates in order to create a sense of mutual respect.

After conducting a factor analysis of the data obtained, it becomes a factor and new indicators that can be used for employee performance measurement model in the implementation of BPMS (Business Process Management

System). The new indicators will be used to evaluate the implementation of BPMS on the CV. Macro Komputrindo. The following is a description of factors and indicators of the newly formed:

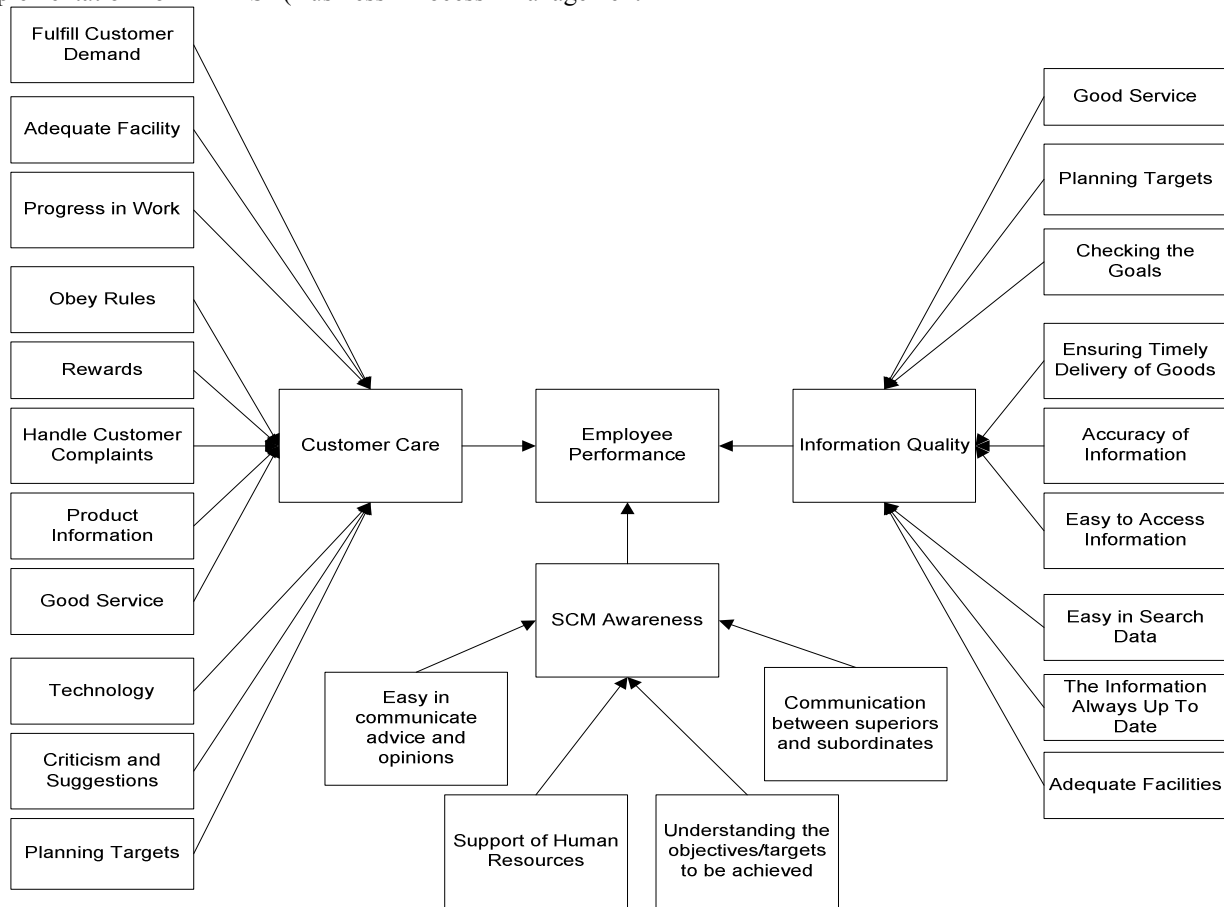


Fig. 2. New Indicators and Factor of Customer Care sufficient to Improve Employee Performance.

Models that can be used as Business Process Management System implementation model known to have 3 new factors that can affect employee performance improvement in using BPMS that is customer care, information quality, and SCM awareness.

From the results of the regression analysis conducted by researchers, the obtained equation can be used to increase employee performance measurement model CV. Macro Komputrindo of implementing Business Process Management System.

$$Y = 4.692 - 0.068X_1 + 0.219X_2 + 0.197X_3$$

From the result of the regression analysis where the first factor is negative, so the first factor which was originally called customer care is because it has a negative value and is changed to Customer Uncared (X_1), the second factor is Information Quality (X_2), the third factor is SCM Awareness (X_3). More detail can be seen in Figure 4 below:

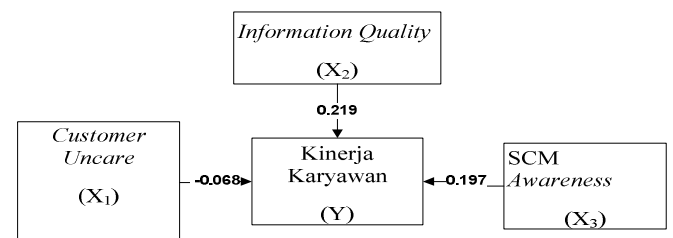


Fig. 3. Factor and values can be used for employee performance measurement model.

Obtained from the above equation, each new factor has a minimum and maximum thresholds as follows:

$$-3.55190 \leq X_1 \leq 1.66998$$

$$-2.85852 \leq X_2 \leq 2.15554$$

$$-4.58847 \leq X_3 \leq 2.15442$$

Of the minimum and maximum limit values of each factor, the simulation can be made to the following conditions:

Kondisi	X1	X2	X3	Constant	Y
	-0.068	0.219	0.197		
Normal	0	0	0	4.692	4.692
Maksimum	1.66998	2.15554	2.15442	4.692	5.474925
Minimum	-3.5519	-2.85852	-4.58847	4.692	3.403585
Ideal	-3.5519	2.15554	2.15442	4.692	5.830013
Ekstrim	1.66998	-2.85852	-4.58847	4.692	3.048497

Fig. 4. Simulation condition factor and the value that can be used for employee performance measurement model.
Source: Author Analysis, 2014

Based on the simulation of the above conditions, the condition of the company can be said to be ideal if the employees of the company can reduce or minimize the level of indifference to customer requests (customer uncared), improving the quality of information (information quality), and increase employee understanding of the process of enterprise SCM (SCM Awareness). Conversely condition of the company could be in an extreme position if the level of employee indifference to customer requests (customer uncared) higher, the quality of the information that is in decline, and employee understanding of the process of enterprise SCM (SCM Awareness) also decreased.

Managerial Implications

From these results, it produces three (3) new factors that may be implicated for the managerial CV. Macro Komputrindo to pay more attention to these three factors in order to improve the performance of employees in the implementation of Business Process Management System. The third factor to consider as follows:

1. The first factor is related to the customer uncared. If employees have a sense of indifference to the wishes/consumer demand will result in a lack of consumer confidence, consumer dissatisfaction with the services of the company, and a decrease in the level of corporate earnings. Therefore, this study gives advice recommendations to companies to increase employee awareness of or attention to the needs of consumers so that consumers can put his trust when shopping, customer satisfaction with services increased, and certainly will increase the company's revenue.
2. The second factor is related to information quality greatly affect the performance of the employee if the employee due to difficulty accessing the information and the information stored is not up to date will have an impact on the error in the delivery of information to consumers. Therefore, to improve these two factors necessary to increase the quality of the information stored and accessible information in real time in order to provide correct information related to products marketed or sold.
3. The third factor is related to SCM awareness greatly affect the performance of employees because employees need to

know and understand what the relevant target or targets to be achieved and how to achieve company goals and objectives. Therefore, employees are required to be able to understand what the target or targets to be achieved, the existence of communication between superiors and subordinates related way as what to do to achieve the desired target, and provide support for the company to achieve its goals that have been determined.

IV. CONCLUSION

Based on the research that has been done, it can be seen:

1. To evaluate the implementation of Business Process Management System (BPMS) which focuses on improving the performance of employees in the use of BPMS.
2. Provide an assessment of the performance and effectiveness of BPM related system development SCM (BPMS).
3. Formed three new factors based on the grouping of a number of indicators that can be used to measure the performance of employees. Where these three factors among them is the first factor (Customer Uncared) formed of an indicator the company's ability to fulfill customer demand, availability of facilities and infrastructure that support, the ability of employees to achieve progress in the work, employees comply with company regulations, the grant awards for achievements, the ability of employees to respond to all customer complaints, employee's ability to provide information to customers, employees provide good service to customers, the support of technology, companies can receive feedback, criticism and suggestions from employees, always held a meeting to plan the targets to be achieved. The second factor (Information Quality) formed from the indicator of the ability of employees to provide good service to customers, always held a meeting to plan the targets to be achieved, the checks of goods, ensure the delivery of goods to customers on time, employees are able to provide accurate information, ease of employees in accessing information, the ease of data retrieval, information about products that are up to date and the presence of adequate facilities. The third factor (SCM Awareness) was formed on the indicator of the opportunity for employees submit suggestions/opinions, the support of human resources, all employees know and understand the objectives/targets to be achieved, and good communication between superiors and subordinates.

The establishment of performance measurement models ideal new employee can be measured as the following equation:

$$Y = 4.692 - 0.068X_1 + 0.219X_2 + 0.197X_3$$

With minimum and maximum thresholds as follows:

$$-3.55190 \leq X_1 \leq 1.66998$$

$$-2.85852 \leq X_2 \leq 2.15554$$

$$-4.58847 \leq X_3 \leq 2.15442$$

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